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Inventor(s)	Han; Junghoon et al.

Organometallic compound and organic light-emitting device including the same

Abstract

Provided are an organometallic compound and an organic light-emitting device including the same. An organic layer of the organic light-emitting device includes an organometallic compound represented by Formula 1: ##STR00001##

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) Korean Patent Application No. 10-2018-0163306, filed on Dec. 17, 2018, in the Korean Intellectual Property Office, and entitled: “Organometallic Compound and Organic Light-Emitting Device Including the Same,” is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) Embodiments relate to an organometallic compound and an organic light-emitting device including the same.

2. Description of the Related Art

(3) Organic light-emitting devices are self-emission devices that produce full-color images, and also have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of brightness, driving voltage, and response speed, compared to devices in the art.

(4) An example of such organic light-emitting devices may include a first electrode disposed on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode, which are sequentially disposed on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transit from an excited state to a ground state, thereby generating light.

SUMMARY

(5) Embodiments are directed to an organic light-emitting device, including a first electrode, a second electrode, and an organic layer between the first electrode and the second electrode and including an emission layer, the organic layer including an organometallic compound represented by Formula 1:

(6) ##STR00002## wherein, in Formula 1, M.sub.11 may be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm), Y.sub.10, Y.sub.20, Y.sub.30, and Y.sub.40 may each independently be N or C, Y.sub.11, Y.sub.12, Y.sub.21, Y.sub.22, Y.sub.31, Y.sub.32, Y.sub.41, and Y.sub.42 may each independently be N or C, A.sub.10,

A.sub.20, A.sub.30, and A.sub.40 may each independently be a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, T.sub.11 to T.sub.14 may each independently be a single bond, $*-O-*$, or $*-S-*$, L.sub.11 may be a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, $*-O-*$, $*-S-*$, $*-C(R.sub.1)(R.sub.2)-*$, $*-C(R.sub.1)=*$, $*=C(R.sub.1)-*$, $*-C(R.sub.1)=C(R.sub.2)-*$, $*-C(=S)-*$, $*-B(R.sub.1)-*$, $*-N(R.sub.1)-*$, $*-P(R.sub.1)-*$, $*-Si(R.sub.1)(R.sub.2)-*$, $*-P(R.sub.1)(R.sub.2)-*$, or $*-Ge(R.sub.1)(R.sub.2)-*$, a₁₁ may be 1, 2, 3, 4, or 5, at least one of L.sub.11(s) in the number of a₁₁ may be a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, L.sub.12 to L.sub.14 may each independently be a single bond, $*-O-*$, $*-S-*$, $*-C(R.sub.1)(R.sub.2)-*$, $*-C(R.sub.1)=*$, $*=C(R.sub.1)-*$, $*-C(R.sub.1)=C(R.sub.2)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R.sub.1)-*$, $*-N(R.sub.1)-*$, $*-P(R.sub.1)-*$, $*-Si(R.sub.1)(R.sub.2)-*$, $*-P(R.sub.1)(R.sub.2)-*$, or $*-Ge(R.sub.1)(R.sub.2)-*$, a₁₂, a₁₃, and a₁₄ may each independently be 0, 1, 2, 3, 4, or 5, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —N(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), or —P(=S)(Q.sub.1)(Q.sub.2), R.sub.1 and R.sub.2 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, R.sub.1 and R.sub.10; R.sub.1 and R.sub.20; R.sub.1 and R.sub.30; or R.sub.1 and R.sub.40 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b₁₀, b₂₀, b₃₀, and b₄₀ may each independently be 1, 2, 3, 4, 5, 6, 7, or 8, * and *' each indicate a binding site to a neighboring atom, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino

group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), or —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), or —P(=O)(Q.sub.21)(Q.sub.22); or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C.sub.1-C.sub.60 alkyl group substituted with deuterium, —F, or a cyano group, a C.sub.6-C.sub.60 aryl group substituted with deuterium, —F, or a cyano group, a biphenyl group, or a terphenyl group.

(7) Embodiments are also directed to an organometallic compound represented by Formula 1.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with FIGURE which is a schematic view of an organic light-emitting device according to an example embodiment.

DETAILED DESCRIPTION

(2) Example embodiments will now be described more fully hereinafter with reference to the

accompanying drawing; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey exemplary implementations to those skilled in the art.

(3) In the drawing figure, the dimensions of layers and regions may be exaggerated for clarity of illustration. Like reference numerals refer to like elements throughout.

(4) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(5) An organometallic compound according to an example embodiment is represented by Formula 1:

(6) ##STR00003##

(7) In Formula 1, M.sub.11 may be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm).

(8) In an example embodiment, M.sub.11 may be Pt, Pd, Cu, Ag, or Au.

(9) For example, M.sub.11 may be Pt.

(10) In Formula 1, Y.sub.10, Y.sub.20, Y.sub.30, and Y.sub.40 may each independently be N or C.

(11) In an example embodiment, in Formula 1, Y.sub.10, Y.sub.20, and Y.sub.30 may each be C, and Y.sub.40 may be N, Y.sub.10, Y.sub.20, and Y.sub.40 may each be C, and Y.sub.30 may be N, Y.sub.10, Y.sub.30, and Y.sub.40 may each be C, and Y.sub.20 may be N, Y.sub.20, Y.sub.30, and Y.sub.40 may each be C, and Y.sub.10 may be N, Y.sub.10 and Y.sub.40 may each be C, and Y.sub.20 and Y.sub.30 may each be N, Y.sub.10 and Y.sub.40 may each be N, and Y.sub.20 and Y.sub.30 may each be C, Y.sub.10 and Y.sub.20 may each be C, and Y.sub.30 and Y.sub.40 may each be N, Y.sub.10 and Y.sub.20 may each be N, and Y.sub.30 and Y.sub.40 may each be C, Y.sub.10 and Y.sub.30 may each be C, and Y.sub.20 and Y.sub.40 may each be N, or Y.sub.10 and Y.sub.30 may each be N, and Y.sub.20 and Y.sub.40 may each be C.

(12) In an example embodiment, at least one of Y.sub.10 and Y.sub.20 may be C, (i) when Y.sub.10 is C, a bond between Y.sub.10 and T.sub.11 or a bond between Y.sub.10 and M.sub.11 may be a coordinate bond, and (ii) when Y.sub.20 is C, a bond between Y.sub.20 and T.sub.12 or a bond between Y.sub.20 and M.sub.11 may be a coordinate bond.

(13) In an example embodiment, Y.sub.30 and Y.sub.40 may each be C.

(14) In Formula 1, A.sub.10, A.sub.20, A.sub.30, and A.sub.40 may each independently be a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.

(15) In an example embodiment, A.sub.10 to A.sub.40 may each independently be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an indeno pyridine group, an indolopyridine group, a benzofuopyridine group, a benzothienopyridine group, a benzosilolopyridine group, an indeno pyrimidine group, an indolopyrimidine group, a benzofuopyrimidine group, a benzothienopyrimidine group, a benzosilolopyrimidine group, a dihydropyridine group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, an imidazopyrimidine group, a 2,3-

dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

(16) In an example embodiment, at least one of A.sub.10 and A.sub.20 may be an N-containing C.sub.1-C.sub.60 heterocyclic group.

(17) In an example embodiment, A.sub.10 and A.sub.20 may each independently be an N-containing C.sub.1-C.sub.60 heterocyclic group.

(18) In an example embodiment, A.sub.10 and A.sub.20 may each independently be a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

(19) In an example embodiment, at least one of A.sub.10 and A.sub.20 may be an N-containing 5-membered heterocyclic group or an N-containing 7-membered heterocyclic group, or at least one of A.sub.10 and A.sub.20 may be condensed with a 6-membered carbocyclic group or a 6-membered heterocyclic group.

(20) In an example embodiment, A.sub.10 and A.sub.20 may each independently be an N-containing 5-membered heterocyclic group or an N-containing 7-membered heterocyclic group, and A.sub.10 and A.sub.20 may be condensed with a 6-membered carbocyclic group or a 6-membered heterocyclic group.

(21) In an example embodiment, A.sub.10 and A.sub.20 may each independently be a group represented by one of Formulae 10-1 to 10-3:

(22) ##STR00004##

(23) In Formulae 10-1 to 10-3, Y.sub.13 to Y.sub.18 and X.sub.11 to X.sub.14 may each independently be N or C(R.sub.10), R.sub.10 is the same as described above, and *, *, and *' each indicate a binding site to a neighboring atom.

(24) In an example embodiment, A.sub.30 and A.sub.40 may each independently be a group represented by one of Formulae 2-1 to 2-43:

(25) ##STR00005## ##STR00006## ##STR00007## ##STR00008## ##STR00009##

(26) In Formulae 2-1 to 2-43, X.sub.21 to X.sub.23 may each independently be C(R.sub.24) or C—* and at least two of X.sub.21 to X.sub.23 may be C—*, X.sub.24 may be N—*. X.sub.25 and X.sub.26 may each independently be C(R.sub.24) or C—*, and at least one of X.sub.25 and X.sub.26 may be C—*, X.sub.27 and X.sub.28 may each independently be N, N(R.sub.25), or N—*, and X.sub.29 may be C(R.sub.24) or C—*, wherein i) at least one of X.sub.27 and X.sub.28 may be N—*, and X.sub.29 may be C—*, or ii) X.sub.27 and X.sub.28 may each be N—*, and X.sub.29 may be C(R.sub.24), R.sub.21 to R.sub.25 may each independently be the same as described in connection with R.sub.10, b21 may be 1, 2, or 3, b22 may be 1, 2, 3, 4, or 5, b23 may be 1, 2, 3, or 4, b24 may be 1 or 2, and * indicates a binding site to a neighboring atom.

(27) In an example embodiment, in Formula 1, A.sub.30 and A.sub.40 may each independently be a group represented by one of Formulae 2-1 to 2-9.

(28) In an example embodiment, in Formula 1, A.sub.30 and A.sub.40 may each independently be a benzene group, a pyridine group, a pyrimidine group, or a triazine group.

(29) In Formula 1, T.sub.11 to T.sub.14 may each independently be a single bond, *—O—*, or *—S—*.

- (30) In an example embodiment, in Formula 1, each of T.sub.11 to T.sub.14 may be a single bond; T.sub.11 may be O or S, and T.sub.12 to T.sub.14 may each be a single bond; T.sub.12 may be O or S, and T.sub.11, T.sub.13, and T.sub.14 may each be a single bond; T.sub.13 may be O or S, and T.sub.11, T.sub.12, and T.sub.14 may each be a single bond; or T.sub.14 may be O or S, and T.sub.11, T.sub.12, and T.sub.13 may each be a single bond.
- (31) For example, in Formula 1, each of T.sub.11 to T.sub.14 may each be a single bond.
- (32) In an example embodiment, a bond between Y.sub.10 and T.sub.11 or a bond between Y.sub.10 and M.sub.11 may be a coordinate bond.
- (33) In an example embodiment, a bond between Y.sub.20 and T.sub.12 or a bond between Y.sub.20 and M.sub.11 may be a coordinate bond.
- (34) In an example embodiment, a bond between Y.sub.30 and T.sub.13 or a bond between Y.sub.30 and M.sub.11 may be a covalent bond.
- (35) In an example embodiment, a bond between Y.sub.40 and T.sub.14 or a bond between Y.sub.40 and M.sub.11 may be a covalent bond.
- (36) In Formula 1, L.sub.11 may be a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, or *—Ge(R.sub.1)(R.sub.2)—*, a₁₁ may be 1, 2, 3, 4, or 5, and at least one of L.sub.11(s) in the number of a₁₁ may be a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group.
- (37) In an example embodiment, at least one of L.sub.11(s) in the number of a₁₁ may be: a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group; or a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group.
- (38) In an example embodiment, at least one of L.sub.11(s) in the number of a₁₁ may be a group represented by one of Formulae 3-1 to 3-5:
- (39) ##STR00010##
- (40) In Formulae 3-1 to 3-5, R.sub.3 may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, b₄ may be an integer from 1 to 4, b₆ may be an integer from 1 to 6, b₈ may be an integer from 1 to 8, b₁₀ may be an integer from 1 to 10, b₁₂ may be an integer from 1 to 12, and * and *' each indicate a binding site to a neighboring atom.
- (41) In an example embodiment, a₁₁ may be 1, and L.sub.11 may be: a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group; or a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino

group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkenyl group, a C.sub.1-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group.

(42) In an example embodiment, a₁₁ may be 1, and L.sub.11 may be a group represented by one of Formulae 3-1 to 3-5.

(43) In Formula 1, L.sub.12 to L.sub.14 may each independently be a single bond, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, or *—Ge(R.sub.1)(R.sub.2)—*.

(44) In an example embodiment, L.sub.12 to L.sub.14 may each independently be a single bond, *—O—*, *—S—*, *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, *—Si(R.sub.1)(R.sub.2)—*, or *—B(R.sub.1)—*.

(45) In an example embodiment, in Formula 1, L.sub.12 may be *—C(R.sub.1)(R.sub.2)—* or *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.2 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, L.sub.12 may be *—C(R.sub.1)(R.sub.2)—* for *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.2 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, or L.sub.14 may be *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, or *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.2 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group.

(46) In an example embodiment, in Formula 1, L.sub.12 may be *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, or *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.2 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, L.sub.12 may be *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, or *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.2; or R.sub.1 and R.sub.30 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, or L.sub.14 may be *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, or *—Si(R.sub.1)(R.sub.2)—*, and R.sub.1 and R.sub.40 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group.

(47) In an example embodiment, in Formula 1, L.sub.12 and L.sub.14 may each be a single bond, and L.sub.13 may be *—O—*, *—S—*, *—N(R.sub.1)—*, *—C(R.sub.1)(R.sub.2)—*, *—Si(R.sub.1)(R.sub.2)—*, or *—B(R.sub.1)—*.

(48) In Formula 1, a₁₂, a₁₃, and a₁₄ may each independently be 0, 1, 2, 3, 4, or 5. When a₁₂ is 0, A.sub.20 and A.sub.30 may not be linked, when a₁₃ is 0, A.sub.30 and A.sub.40 may not be linked, and when a₁₄ is 0, A.sub.30 and A.sub.40 may not be linked.

(49) In an example embodiment, a₁₂, a₁₃, and a₁₄ may each independently be 1, 2, or 3

(50) In Formula 1, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted

C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —N(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), or —P(=S)(Q.sub.1)(Q.sub.2), R.sub.1 and R.sub.10; R.sub.1 and R.sub.20; R.sub.1 and R.sub.30; or R.sub.1 and R.sub.40 may be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b10, b20, b30, and b40 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8, and Q.sub.1 to Q.sub.3 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, or a terphenyl group.

(51) In an example embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, or a biphenyl group; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphtho silolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno

carbazolyl group, or an indolocarbazolyl group; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphthosilolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno carbazolyl group, or an indolocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphthosilolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno carbazolyl group, an indolocarbazolyl

group, —Si(Q.sub.30)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or —P(=S)(Q.sub.31)(Q.sub.32); or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —N(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), or —P(=S)(Q.sub.1)(Q.sub.2), and Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C.sub.1-C.sub.60 alkyl group substituted with at least deuterium, —F, or a cyano group, a C.sub.6-C.sub.60 aryl group substituted with deuterium, —F, or a cyano group, a biphenyl group, or a terphenyl group.

(52) In an example embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, or a biphenyl group; a group represented by one of Formulae 5-1 to 5-25; or a group represented by one of Formulae 6-1 to 6-55, and R.sub.1 and R.sub.2 may be linked to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthalene group, or a fluorene group,

(53) ##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018##

(54) In Formulae 5-1 to 5-25 and 6-1 to 6-55, Y.sub.31 may be O, S, C(Z.sub.34)(Z.sub.35), N(Z.sub.34), Or Si(Z.sub.34)(Z.sub.35), Z.sub.31 to Z.sub.35 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, or a triazinyl group, e2 may be 1 or 2, e3 may be an integer from 1 to 3, e4 may be an integer from 1 to 4, e5 may be an integer from 1 to 5, e6 may be an integer from 1 to 6, e7 may be an integer from 1 to 7, e9 may be an integer from 1 to 9, and * indicates a binding site to a neighboring atom.

(55) In an example embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, or a biphenyl group; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group; or a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an

acridinyl group, a dibenzofuran group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, or a biphenyl group, and R.sub.1 and R.sub.2 may be linked to form a cyclopentane group, a cyclohexane group, or a fluorene group.

(56) In an example embodiment, the organometallic compound may be one of Compounds 1 to 40:

(57) ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026##

(58) In an example embodiment, the organometallic compound represented by Formula 1 has a structure in which at least one of L.sub.11(s) in the number of a11 is a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group. Without being bound by theory, it is believed that, since the cycloalkylene group is positioned outside a plane (in which a metal M.sub.11 and rings A.sub.11 to A.sub.14 are included) to form a three-dimensional structure, interaction with other molecules may be suppressed, and the formation of excimer and exciplex may be inhibited. Therefore, when the organometallic compound represented by Formula 1 according to the present example embodiment is applied to the organic light-emitting device, it may be advantageous to energy transfer, luminance efficiency may be high, high color purity may be realized, and lifespan characteristics may be improved.

(59) In addition, when ring A.sub.10 in the organometallic compound represented by Formula 1 is carbene, high color purity and excellent lifespan characteristics may be provided due to strong σ -donation and π -backdonation effects.

(60) In an example embodiment, the organometallic compound represented by Formula 1 includes a moiety providing a rigid structure. Thus, an organic light-emitting device including the organometallic compound may have high durability and a long lifespan.

(61) The organometallic compound may emit blue light. For example, the organometallic compound may emit blue light (bottom emission CIE.sub.x,y color coordinates 0.135, 0.15 to 0.30) having a wavelength of maximum emission of about 440 nm to about 500 nm, for example, a wavelength of maximum emission of about 460 nm to about 495 nm. Therefore, the organometallic compound represented by Formula 1 may be usefully used to manufacture an organic light-emitting device that emits blue light.

(62) A synthesis method for the organometallic compound represented by Formula 1 will be apparent to those of skill in the art by referring to the examples below.

(63) At least one of the organometallic compound of Formula 1 may be used between a pair of electrodes of an organic light-emitting device. For example, the organometallic compound may be included in an emission layer. The organometallic compound may act as a dopant in the emission layer.

(64) According to an example embodiment, an organic light-emitting device includes a first electrode, a second electrode facing the first electrode, and an organic layer between the first electrode and the second electrode. The organic layer may include an emission layer. The organic layer may include at least one organometallic compound represented by Formula 1.

(65) The expression “(an organic layer) includes at least one organometallic compound” as used herein may include a case in which “(an organic layer) includes identical compounds represented by Formula 1” and a case in which “(an organic layer) includes two or more different organometallic compounds represented by Formula 1.”

(66) For example, the organic layer may include, as the organometallic compound, only Compound 1. In this regard, Compound 1 may exist in an emission layer of the organic light-emitting device. In an example embodiment, the organic layer may include, as the organometallic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may

exist in an electron transport region).

(67) In an example embodiment, the first electrode of the organic light-emitting device may be an anode, the second electrode of the organic light-emitting device may be a cathode, the organic layer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or a combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

(68) The term “organic layer” as used herein refers to a single layer and/or a plurality of layers disposed between the first electrode and the second electrode of the organic light-emitting device. A material included in the “organic layer” is not limited to an organic material.

(69) In an example embodiment, the emission layer may include an organometallic compound represented by Formula 1, and may further include a host, wherein an amount of the host in the emission layer may be greater than that of the organometallic compound in the emission layer.

(70) In an example embodiment, the emission layer may further include a host, and the host may include a silyl-containing compound, a phosphine oxide-containing compound, or a combination thereof.

(71) In an example embodiment, the hole transport region may include an electron blocking layer, and the electron blocking layer may include the organometallic compound; or the electron transport region may include a hole blocking layer, and the hole blocking layer may include the organometallic compound.

(72) In an example embodiment, the hole transport region may include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level of about -3.5 eV or less.

(73) In an example embodiment, the electron transport region may include a hole blocking layer, and the hole blocking layer may directly contact the emission layer, and the hole blocking layer may include a phosphine oxide-containing compound or a silyl-containing compound.

DESCRIPTION OF THE FIGURE

(74) The FIGURE is a schematic view of an organic light-emitting device **10** according to an example embodiment. The organic light-emitting device **10** includes a first electrode **110**, an organic layer **150**, and a second electrode **190**.

(75) Hereinafter, the structure of the organic light-emitting device **10** according to an example embodiment and a method of manufacturing the organic light-emitting device **10** will be described in connection with the FIGURE.

(76) [First Electrode **110**]

(77) In the FIGURE, a substrate may be additionally disposed under the first electrode **110** or above the second electrode **190**. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

(78) The first electrode **110** may be formed by depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, the material for a first electrode may be selected from materials with a high work function to facilitate hole injection.

(79) The first electrode **110** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming a first electrode may be selected from indium tin oxide (ITO), indium zinc oxide (ILO), tin oxide (SnO.sub.2), zinc oxide (ZnO), and a combinations thereof. In an example embodiment, when the first electrode **110** is a semi-transmissive electrode or a reflective electrode, a material for forming a first electrode may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and a combinations thereof.

(80) The first electrode **110** may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO.

(81) [Organic Layer **150**]

(82) The organic layer **150** is disposed on the first electrode **110**. The organic layer **150** may include an emission layer.

(83) The organic layer **150** may further include a hole transport region between the first electrode **110** and the emission layer, and an electron transport region between the emission layer and the second electrode **190**.

(84) [Hole Transport Region in Organic Layer **150**]

(85) The hole transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(86) The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

(87) For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers may be sequentially stacked from the first electrode **110** in this stated order.

(88) The hole transport region may include at least one selected from m-MTDATA, TDATA, 2-TNATA, NPB(NPD), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTDP, 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, and a compound represented by Formula 202:

(89) ##STR00027## ##STR00028## ##STR00029## ##STR00030##

(90) In Formulae 201 and 202, L.sub.201 to L.sub.204 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, L.sub.205 may be selected from *—O—*, *—S—*, *—N(Q.sub.201)—*, a substituted or unsubstituted C.sub.1-C.sub.20 alkylene group, a substituted or unsubstituted C.sub.2-C.sub.20 alkenylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xa1 to xa4 may each independently be an integer of 0 to 3, xa5 may be an integer of 1 to 10, and R.sub.201 to R.sub.204 and Q.sub.201 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.60 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10

heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

(91) For example, in Formula 202, R.sub.201 and R.sub.202 may be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group, and R.sub.203 and R.sub.204 may be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

(92) In an example embodiment, in Formulae 201 and 202, L.sub.201 to L.sub.205 may each independently be selected from: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthenylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthenylene group, a fluorenylenylene group, a spiro-bifluorenylenylene group, a benzofluorenylenylene group, a dibenzofluorenylenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(93) In an example embodiment, xa1 to xa4 may each independently be 0, 1, or 2.

(94) In an example embodiment, xa5 may be 1, 2, 3, or 4.

(95) In an example embodiment, R.sub.201 to R.sub.204 and Q.sub.201 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described above.

(96) In an example embodiment, in Formula 201, at least one selected from R.sub.201 to R.sub.203 may each independently be selected from: a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

(97) In an example embodiment, in Formula 202, i) R.sub.201 and R.sub.202 may be linked via a single bond, and/or ii) R.sub.203 and R.sub.204 may be linked via a single bond.

(98) In an example embodiment, in Formula 202, at least one selected from R.sub.201 to R.sub.204 may each independently selected from: a carbazolyl group; and a carbazolyl group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

(99) The compound represented by Formula 201 may be represented by Formula 201A:

(100) ##STR00031##

(101) For example, the compound represented by Formula 201 may be represented by Formula 201A(1):

(102) ##STR00032##

(103) For example, the compound represented by Formula 201 may be represented by Formula 201A-1:

(104) ##STR00033##

(105) For example, the compound represented by Formula 202 may be represented by Formula 202A:

(106) ##STR00034##

(107) For example, the compound represented by Formula 202 may be represented by Formula 202A-1:

(108) ##STR00035##

(109) In Formulae 201A, 201A(1), 201A-1, 202A, and 202A-1, L.sub.201 to L.sub.203, xa1 to xa3, xa5 and R.sub.202, to R.sub.204 are the same as described above, R.sub.211 and R.sub.212 may each independently be the same as defined in connection with R.sub.203, and R.sub.213 to R.sub.217 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a Spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(110) The hole transport region may include at least one compound selected from Compounds HT1 to HT39:

(111) ##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040##

##STR00041## ##STR00042##

(112) A thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example about

100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

(113) The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described above.

(114) [p-Dopant]

(115) The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

(116) The charge-generation material may be, for example, a p-dopant.

(117) In an example embodiment, the p-dopant may have a LUMO energy level of about -3.5 eV or less.

(118) The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound.

(119) For example, the p-dopant may include at least one selected from: a quinone derivative, such as tetracyanoquinodimethane (TCNQ) or 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ); a metal oxide, such as tungsten oxide or molybdenum oxide; 1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and a compound represented by Formula 221:

(120) ##STR00043##

(121) In Formula 221, R.sub.221 to R.sub.223 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R.sub.221 to R.sub.223 may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C.sub.1-C.sub.20 alkyl group substituted with —F, a C.sub.1-C.sub.20 alkyl group substituted with —Cl, a C.sub.1-C.sub.20 alkyl group substituted with —Br, and a C.sub.1-C.sub.20 alkyl group substituted with —I.

[Emission Layer in Organic Layer 150]

(122) When the organic light-emitting device 10 is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a sub-pixel. In an example embodiment, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In an example embodiment, the emission layer may include two or more materials selected from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

(123) The emission layer may include a host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant. The phosphorescent dopant may include the organometallic compound represented by Formula 1:

(124) In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host.

(125) A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for

example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

(126) [Host in Emission Layer]

(127) In an example embodiment, the host may include a compound represented by Formula 301:
[Ar.sub.301].sub.xb11-[(L.sub.301).sub.xb1-R.sub.301].sub.xb21 <Formula 301>

(128) In Formula 301, Ar.sub.301 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xb11 may be 1, 2, or 3, L.sub.301 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xb1 may be an integer of 0 to 5, R.sub.301 may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), and —P(=O)(Q.sub.301)(Q.sub.302), xb21 may be an integer of 1 to 5, Q.sub.301 to Q.sub.303 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(129) In an example embodiment, Ar.sub.301 in Formula 301 may be selected from: a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(130) In Formula 301, when xb11 is two or more, two or more Ar.sub.301(s) may be linked via a

single bond.

(131) In an example embodiment, the compound represented by Formula 301 may be represented by Formula 301-1 or 301-2:

(132) ##STR00044##

(133) In Formulae 301-1 to 301-2, A.sub.301 to A.sub.304 may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a pyridine group, a pyrimidine group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, and a dinaphthothiophene group, X.sub.301 may be O, S, or N-[(L.sub.304).sub.xb4-R.sub.304], R.sub.311 to R.sub.314 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), xb22 and xb23 may each independently be 0, 1, or 2, L.sub.301, xb1, R.sub.301, and Q.sub.31 to Q.sub.33 are the same as described above, L.sub.302 to L.sub.304 may each independently be the same as defined in connection with L.sub.301, xb2 to xb4 may each independently be the same as defined in connection with xb1, and R.sub.302 to R.sub.304 may each independently be the same as defined in connection with R.sub.301.

(134) For example, in Formulae 301, 301-1, and 301-2, L.sub.31 to L.sub.304 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a

pyridinylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a thiadiazolylene group, an oxadiazolylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described above.

(135) In an example embodiment, in Formulae 301, 301-1, and 301-2, R.sub.301 to R.sub.304 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an

indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described above.

(136) In an example embodiment, the host may include an alkaline earth metal complex. For example, the host may be selected from a Be complex (for example, Compound H55), a Mg complex, and a Zn complex.

(137) The host may include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55:

(138) ##STR00045## ##STR00046## ##STR00047## ##STR00048## ##STR00049##
##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055##
##STR00056##

(139) In an example embodiment, the host may include at least one selected from a silicon-containing compound (for example, BCPDS in the following examples) and a phosphine oxide-containing compound (for example, POPCPA in the following examples).

(140) In an example embodiment, the host may include only one compound, or two or more different compounds (for example, a host used in the following examples includes BCPDS and POPCPA).

(141) [Phosphorescent Dopant Included in Emission Layer in Organic Layer **150**]

(142) The phosphorescent dopant may include an organometallic compound represented by Formula 1:

(143) The phosphorescent dopant may include an organometallic complex represented by Formula 401:

(144) ##STR00057##

(145) In Formulae 401 and 402, M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm), L.sub.401 may be selected from ligands represented by Formula 402, and xc1 may be 1, 2, or 3, wherein, when xc1 is two or more, two or more L.sub.401(s) may be identical to or different from each other, L.sub.402 may be an organic ligand, and xc2 may be an integer of 0 to 4, wherein, when xc2 is two or more, two or more L.sub.402(5) may be identical to or different from each other, X.sub.401 to X.sub.404 may each independently be nitrogen or carbon, X.sub.401 and X.sub.403 may be linked via a single bond or a double bond, and X.sub.402 and X.sub.404 may be linked via a single bond or a double bond, A.sub.401 and A.sub.402 may each independently be selected from a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, X.sub.405 may be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.411)—*, *—C(Q.sub.411)(Q.sub.412)—*, *—C(Q.sub.411)=C(Q.sub.412)—*, *—C(Q.sub.411)=*, or *=C=*, wherein Q.sub.411 and Q.sub.412 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, X.sub.406 may be a single bond, O, or S, R.sub.401 and R.sub.402 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.20 alkyl group, a substituted or unsubstituted C.sub.1-C.sub.20 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.6-C.sub.60 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), wherein Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a C.sub.6-C.sub.20 aryl group, and a C.sub.1-C.sub.20 heteroaryl group, xc11 and xc12 may each independently be an integer of 0 to 10, and in Formula 402, * and *' each indicate a binding site to M in Formula 401.

(146) In an example embodiment, in Formula 402, A.sub.401 and A.sub.402 may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, an indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

(147) In an example embodiment, in Formula 402, i) X.sub.401 may be nitrogen, and X.sub.402 may be carbon, or ii) X.sub.401 and X.sub.402 may each be nitrogen at the same time.

(148) In an example embodiment, R.sub.401 and R.sub.402 in Formula 402 may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-

C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and

(149) —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), and Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group.

(150) In an example embodiment, when xc1 in Formula 401 is two or more, two A.sub.401(s) in two or more L.sub.401(s) may be linked via X.sub.407, which is a linking group, or two A.sub.402(s) in two or more L.sub.401(s) may be linked via X.sub.408, which is a linking group (see Compounds PD1 to PD4 and PD7). X.sub.407 and X.sub.408 may each independently be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.413)—*, *—C(Q.sub.413)(Q.sub.414)—*, or *—C(Q.sub.413)=C(Q.sub.414)—* (wherein Q.sub.413 and Q.sub.414 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group).

(151) L.sub.402 in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For example, L.sub.402 may be selected from halogen, diketone (for example, acetylacetonate), carboxylic acid (for example, picolinate), —C(=O), isonitrile, —CN, and phosphorus (for example, phosphine, or phosphite).

(152) In an example embodiment, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25:

(153) ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062##
[Fluorescent Dopant in Emission Layer]

(154) The fluorescent dopant may further include an arylamine compound or a styrylamine compound.

(155) The fluorescent dopant may include a compound represented by Formula 501:

(156) ##STR00063##

(157) In Formula 501, Ar.sub.501 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, L.sub.501 to L.sub.503 may each independently be selected from a substituted or unsubstituted C.sub.3-

C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xd1 to xd3 may each independently be an integer of 0 to 3; R.sub.501 and R.sub.502 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, xd4 may be an integer of 1 to 6.

(158) In an example embodiment, Ar.sub.501 in Formula 501 may be selected from: a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group; and a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(159) In an example embodiment, L.sub.501 to L.sub.503 in Formula 501 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a

dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(160) In an example embodiment, R.sub.501 and R.sub.502 in Formula 501 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(161) In an example embodiment, xd4 in Formula 501 may be 2.

(162) For example, the fluorescent dopant may be selected from Compounds FD1 to FD22:

(163) ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068##
##STR00069##

(164) In an example embodiment, the fluorescent dopant may be selected from the following compounds.

(165) ##STR00070##

[Electron Transport Region in Organic Layer **150**]

(166) The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(167) The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer.

(168) For example, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer

structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers may be sequentially stacked from an emission layer.

(169) The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

(170) The “ π electron-depleted nitrogen-containing ring” indicates a C.sub.1-C.sub.60 heterocyclic group having at least one *—N=* moiety as a ring-forming moiety.

(171) For example, the “ π electron-depleted nitrogen-containing ring” may be i) a 5-membered to 7-membered heteromonocyclic group having at least one *—N=* moiety, ii) a heteropolycyclic group in which two or more 5-membered to 7-membered heteromonocyclic groups each having at least one *—N=* moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to 7-membered heteromonocyclic groups, each having at least one *—N=* moiety, is condensed with at least one C.sub.5-C.sub.60 carbocyclic group.

(172) Examples of the π electron-depleted nitrogen-containing ring include an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazol group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group.

(173) For example, the electron transport region may include a compound represented by Formula 601:

[Ar.sub.601].sub.xe11-[(L.sub.601).sub.xe1-R.sub.601].sub.xe21 <Formula 601>

(174) In Formula 601, Ar.sub.601 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xe11 may be 1, 2, or 3, L.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group; xe1 may be an integer of 0 to 5, R.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $\text{—Si(Q.sub.601)(Q.sub.602)(Q.sub.603)}$, —C(=O)(Q.sub.601) , $\text{—S(=O).sub.2(Q.sub.601)}$, and $\text{—P(=O)(Q.sub.601)(Q.sub.602)}$, Q.sub.601 to Q.sub.603 may each independently be a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and xe21 may be an integer of 1 to 5.

(175) In an example embodiment, at least one of Ar.sub.601(s) in the number of xe11 and R.sub.601(s) in the number of xe21 may include the π electron-depleted nitrogen-containing ring.

(176) In an example embodiment, Ar.sub.601 in Formula 601 may be selected from: a benzene

group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group; and a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(177) When xe11 in Formula 601 is two or more, two or more Ar.sub.601(s) may be linked via a single bond.

(178) In an example embodiment, Ar.sub.601 in Formula 601 may be an anthracene group.

(179) In an example embodiment, a compound represented by Formula 601 may be represented by Formula 601-1:

(180) ##STR00071##

(181) In Formula 601-1, X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), and X.sub.616 may be N or C(R.sub.616), wherein at least one selected from X.sub.614 to X.sub.616 may be N, L.sub.611 to L.sub.613 may each independently be the same as defined in connection with L.sub.601, xe611 to xe613 may each independently be the same as defined in connection with xe1, R.sub.611 to R.sub.613 may each independently be the same as defined in connection with R.sub.601, and R.sub.614 to R.sub.616 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(182) In an example embodiment, L.sub.601 and L.sub.611 to L.sub.613 in Formulae 601 and 601-1 may each independently be selected from: a phenylene group, a naphthylene group, a

fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an

imidazopyrimidinyl group, and an azacarbazolyl group.

(183) In an example embodiment, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

(184) In an example embodiment, in Formulae 601 and 601-1, R.sub.601 and R.sub.611 to R.sub.613 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl

group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and —S(=O).sub.2(Q.sub.601) and —P(=O)(Q.sub.601) (Q.sub.602), and Q.sub.601 and Q.sub.602 may be the same as described above.

(185) The electron transport region may include at least one compound selected from Compounds ET1 to ET36:

(186) ##STR00072## ##STR00073## ##STR00074## ##STR00075## ##STR00076##

##STR00077## ##STR00078## ##STR00079## ##STR00080## ##STR00081## ##STR00082##

(187) In an example embodiment, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BAlq, 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-4H-1,2,4-triazole (TAZ), and NTAZ:

(188) ##STR00083##

(189) In an example embodiment, the electron transport region may include a phosphine oxide-containing compound (for example, TSPO1 used in the following examples). In an example embodiment, the phosphine oxide-containing compound may be used in a hole blocking layer in the electron transport region.

(190) A thickness of the buffer layer, the hole blocking layer, or the electron control layer may be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thicknesses of the buffer layer, the hole blocking layer, and the electron control layer are within these ranges, the electron transport region may have excellent hole blocking characteristics or electron control characteristics without a substantial increase in driving voltage.

(191) A thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described above, the electron transport layer may have satisfactory electron transport characteristics without a substantial increase in driving voltage.

(192) The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

(193) The metal-containing material may include at least one selected from alkali metal complex and alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a hydroxy diphenyloxadiazole, a hydroxy diphenylthiadiazol, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene.

(194) For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2.

(195) ##STR00084##

(196) The electron transport region may include an electron injection layer that facilitates injection of electrons from the second electrode **190**. The electron injection layer may directly contact the second electrode **190**.

(197) The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(198) The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound,

an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof.

(199) The alkali metal may be selected from Li, Na, K, Rb, and Cs. In an example embodiment, the alkali metal may be Li, Na, or Cs. In an example embodiment, the alkali metal may be Li or Cs.

(200) The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

(201) The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

(202) The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

(203) The alkali metal compound may be selected from alkali metal oxides, such as Li.sub.2O , Cs.sub.2O , or K.sub.2O , and alkali metal halides, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, or KI. In an example embodiment, the alkali metal compound may be selected from LiF, Li.sub.2O , NaF, LiI, NaI, CsI, and KI.

(204) The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO, SrO, CaO, $\text{Ba.sub.xSr.sub.1-xO}$ ($0 < x < 1$), or $\text{Ba.sub.xCa.sub.1-xO}$ ($0 < x < 1$). In an example embodiment, the alkaline earth-metal compound may be selected from BaO, SrO, and CaO.

(205) The rare earth metal compound may be selected from YbF.sub.3 , ScF.sub.3 , ScO.sub.3 , Y.sub.2O.sub.3 , Ce.sub.2O.sub.3 , GdF.sub.3 , and TbF.sub.3 . In an example embodiment, the rare earth metal compound may be selected from YbF.sub.3 , ScF.sub.3 , TbF.sub.3 , YbI.sub.3 , ScI.sub.3 , and TbI.sub.3 .

(206) The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenyloxazole, hydroxy phenylthiazole, hydroxy diphenyloxadiazole, hydroxy diphenylthiadiazol, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene.

(207) The electron injection layer may include or consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof, as described above. In an example embodiment, the electron injection layer may further include an organic material. When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

(208) A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, the electron injection layer may have satisfactory electron injection characteristics without a substantial increase in driving voltage.

(209) [Second Electrode **190**]

(210) The second electrode **190** may be disposed on the organic layer **150** having such a structure. The second electrode **190** may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode **190** may be selected from metal, an alloy, an electrically conductive compound, and a combination thereof, which have a relatively low work function.

(211) The second electrode **190** may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium

(Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO. The second electrode **190** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

(212) The second electrode **190** may have a single-layered structure, or a multi-layered structure including two or more layers.

(213) Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

(214) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10⁻⁸ torr to about 10⁻³ torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(215) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to about 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(216) [General Definition of Substituents]

(217) The term “C.sub.1-C.sub.60 alkyl group” as used herein refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “C.sub.1-C.sub.60 alkylene group” as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.60 alkyl group.

(218) The term “C.sub.2-C.sub.60 alkenyl group” as used herein refers to a hydrocarbon group having at least one double bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C.sub.2-C.sub.60 alkenylene group” as used herein refers to a divalent group having the same structure as the C.sub.2-C.sub.60 alkenyl group.

(219) The term “C.sub.2-C.sub.60 alkynyl group” as used herein refers to a hydrocarbon group having at least one triple bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term “C.sub.2-C.sub.60 alkynylene group” as used herein refers to a divalent group having the same structure as the C.sub.2-C.sub.60 alkynyl group.

(220) The term “C.sub.1-C.sub.60 alkoxy group” as used herein refers to a monovalent group represented by —OA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

(221) The term “C.sub.3-C.sub.10 cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The term “C.sub.3-C.sub.10 cycloalkylene group” as used herein refers to a divalent group having the same structure as the C.sub.3-C.sub.10 cycloalkyl group.

(222) The term “C.sub.1-C.sub.10 heterocycloalkyl group” as used herein refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkylene group” as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

(223) The term "C.sub.3-C.sub.10 cycloalkenyl group" used herein refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one double bond in the ring thereof and no aromaticity, and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term "C.sub.3-C.sub.10 cycloalkenylene group" as used herein refers to a divalent group having the same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

(224) The term "C.sub.1-C.sub.10 heterocycloalkenyl group" as used herein refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one double bond in its ring. Non-limiting examples of the C.sub.1-C.sub.10 heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term "C.sub.1-C.sub.10 heterocycloalkenylene group" as used herein refers to a divalent group having the same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

(225) The term "C.sub.6-C.sub.60 aryl group" as used herein refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and a C.sub.6-C.sub.60 arylene group used herein refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C.sub.6-C.sub.60 aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the rings may be fused to each other.

(226) The term "C.sub.1-C.sub.60 heteroaryl group" as used herein refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. The term "C.sub.1-C.sub.60 heteroarylene group" as used herein refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C.sub.1-C.sub.60 heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the rings may be condensed with each other.

(227) The term "C.sub.6-C.sub.60 aryloxy group" as used herein refers to —OA.sub.102 (wherein A.sub.102 is the C.sub.6-C.sub.60 aryl group), and a C.sub.6-C.sub.60 arylthio group used herein indicates —SA.sub.103 (wherein A.sub.103 is the C.sub.6-C.sub.60 aryl group).

(228) The term "monovalent non-aromatic condensed polycyclic group" as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with each other, only carbon atoms as ring-forming atoms, and at least one non-aromatic ring in its entire molecular structure. A detailed example of the monovalent non-aromatic condensed polycyclic group is a fluorenyl group. The term "divalent non-aromatic condensed polycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

(229) The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other, at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed heteropolycyclic group is a carbazolyl group. The term "divalent non-aromatic condensed heteropolycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

(230) The term "C.sub.5-C.sub.60 carbocyclic group" as used herein refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The C.sub.5-C.sub.60 carbocyclic group may be an aromatic carbocyclic group or a non-aromatic

carbocyclic group. The C.sub.5-C.sub.60 carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In an example embodiment, depending on the number of substituents connected to the C.sub.5-C.sub.60 carbocyclic group, the C.sub.5-C.sub.60 carbocyclic group may be a trivalent group or a quadrivalent group.

(231) The term “C.sub.1-C.sub.60 heterocyclic group” as used herein refers to a group having the same structure as the C.sub.5-C.sub.60 carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

(232) In the specification, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.10 heterocycloalkylene group, the substituted C.sub.3-C.sub.10 cycloalkenylene group, the substituted C.sub.1-C.sub.10 heterocycloalkenylene group, the substituted C.sub.6-C.sub.60 arylene group, the substituted C.sub.1-C.sub.60 heteroarylene group, the substituted divalent non-aromatic condensed polycyclic group, the substituted divalent non-aromatic condensed heteropolycyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group. each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10

heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C.sub.1-C.sub.60 alkyl group substituted with at least one selected from deuterium, —F, and a cyano group, a C.sub.6-C.sub.60 aryl group substituted with at least one selected from deuterium, —F, and a cyano group, a biphenyl group, and a terphenyl group.

(233) The term “Ph” as used herein represents a phenyl group, the term “Me” as used herein represents a methyl group, the term “Et” as used herein represents an ethyl group, the term “ter-Bu” or “Bu.sup.t,” as used herein, represents a tert-butyl group, and the term “OMe” as used herein represents a methoxy group.

(234) The term “biphenyl group” as used herein refers to a “phenyl group substituted with a phenyl group. The “biphenyl group” is a “substituted phenyl group” having a “C.sub.6-C.sub.60 aryl group” as a substituent.

(235) The term “terphenyl group” as used herein refers to a “phenyl group substituted with a biphenyl group. The “terphenyl group” is a “phenyl group” having, as a substituent, a “C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group.” * and *’ as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

(236) The following Examples and Comparative Examples are provided in order to highlight characteristics of one or more embodiments, but it will be understood that the Examples and Comparative Examples are not to be construed as limiting the scope of the embodiments, nor are the Comparative Examples to be construed as being outside the scope of the embodiments. Further, it will be understood that the embodiments are not limited to the particular details described in the Examples and Comparative Examples.

(237) The expression “B was used instead of A” used in describing Synthesis Examples means that an identical number of molar equivalents of A was used in place of molar equivalents of B.

SYNTHESIS EXAMPLES

Synthesis Example 1

Synthesis of Compound 4

(238) ##STR00085## ##STR00086##

1) Synthesis of Intermediate [4-A]

(239) 9.35 g (50 mmol) of bromoanisole, 3.40 g (50 mmol) of imidazole, 23 g (100 mmol) of potassium triphosphate, 1.83 g (10 mmol) of iodine copper, and 1.17 g (10 mmol) of picolinic acid were added to a reaction container and suspended in 100 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column

chromatography to obtain 6.97 g (40 mmol) of Intermediate [4-A].

(240) 2) Synthesis of Intermediate [4-B]

(241) 6.97 g (40 mmol) of Intermediate [4-A] was suspended in an excess of bromic acid solution. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature and neutralized by adding an appropriate amount of sodium hydrogen carbonate thereto. 300 mL of distilled water was added thereto and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 5.77 g (36 mmol) of Intermediate [4-B].

(242) 3) Synthesis of Intermediate [4-C]

(243) 11.80 g (50 mmol) of 1,3-dibromobenzene, 3.40 g (50 mmol) of imidazole, 23 g (100 mmol) of potassium triphosphate, 1.83 g (10 mmol) of iodine copper, and 1.17 g (10 mmol) of picolinic acid were added to a reaction container and suspended in 100 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 7.81 g (35 mmol) of Intermediate [4-C].

(244) 4) Synthesis of Intermediate [4-D]

(245) 5.77 g (36 mmol) of Intermediate [4-B], 7.81 g (35 mmol) of Intermediate [4-C], 16.1 g (70 mmol) of potassium triphosphate, 1.28 g (7 mmol) of iodine copper, and 0.82 g (7 mmol) of picolinic acid were added to a reaction container and suspended in 70 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.47 g (28 mmol) of Intermediate [4-D].

(246) 5) Synthesis of Intermediate [4-E]

(247) 8.47 g (28 mmol) of Intermediate [4-D] and 1.68 g (140 mmol) of 1,2-diiodocyclohexane were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 12.5 g (19.6 mmol) of Intermediate [4-E].

(248) 6) Synthesis of Intermediate [4-F]

(249) 12.5 g (19.6 mmol) of Intermediate [4-E] and 9.58 g (58.8 mmol) of ammonium hexafluorophosphate were suspended in 50 mL of methyl alcohol and 50 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 9.25 g (13.7 mmol) of Intermediate [4-F].

(250) 7) Synthesis of Compound 4

(251) 9.25 g (13.7 mmol) of Intermediate [4-F], 5.6 g (15 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 3.4 g (41 mmol) of sodium acetate were suspended in 160 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110° C. for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by

column chromatography to obtain 2.37 g (4.1 mmol) of Compound 4.

Synthesis Example 2: Synthesis of Compound 9

(252) ##STR00087## ##STR00088##

1) Synthesis of Intermediate [9-A]

(253) 8.75 g (50 mmol) of 1-bromo-3-fluorobenzene was suspended in 100 mL of tetrahydrofuran and stirred at a temperature of -78°C . 24 mL of n-BuLi solution (2.5 M in hexane) was added to the mixed solution and stirred at the same temperature for 2 hours. After the stirring, 7.74 g (60 mmol) of dimethyldichlorosilane was added thereto and further stirred at room temperature for 24 hours. After the reaction was completed, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 9.93 g (40 mmol) of Intermediate [9-A].

(254) 2) Synthesis of Intermediate [9-B]

(255) 9.93 g (40 mmol) of Intermediate [9-A], 8.17 g (120 mmol) of imidazole, and 36.8 g (160 mmol) of potassium triphosphate were added to a reaction container and suspended in 80 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160°C . for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 9.0 g (26 mmol) of Intermediate [9-B].

(256) 3) Synthesis of Intermediate [9-C]

(257) 9.0 g (26 mmol) of Intermediate [9-B] and 1.56 g (130 mmol) of 1,2-diiodocyclohexane were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110°C . for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 12.4 g (18.2 mmol) of Intermediate [9-C].

(258) 4) Synthesis of Intermediate [9-D]

(259) 12.4 g (18.2 mmol) of Intermediate [9-C] and 8.9 g (54.6 mmol) of ammonium hexafluorophosphate were added to a reaction container and suspended in 50 mL of methyl alcohol and 50 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 8.48 g (11.8 mmol) of Intermediate [9-D].

(260) 5) Synthesis of Compound 9

(261) 8.48 g (11.8 mmol) of Intermediate [9-D], 4.85 g (13 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 2.94 g (35.4 mmol) of sodium acetate were suspended in 140 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110°C . for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 1.11 g (1.8 mmol) of Compound 9.

Synthesis Example 3: Synthesis of Compound 14

(262) ##STR00089## ##STR00090##

1) Synthesis of Intermediate [14-A]

(263) 8.75 g (50 mmol) of 1-bromo-3-fluorobenzene was suspended in 100 mL of tetrahydrofuran and stirred at a temperature of -78°C . 24 mL of n-BuLi solution (2.5 M in hexane) was added to the mixed solution and stirred at the same temperature for 2 hours. After the stirring, 9.51 g (60 mmol) of 3-fluorobenzoyl chloride was added thereto and further stirred at room temperature for 24

hours. After the reaction was completed, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.29 g (38 mmol) of Intermediate [14-A].

(264) 2) Synthesis of Intermediate [14-B]

(265) 10.7 g (45.6 mmol) of 2-bromobiphenyl, 11.1 g (45.6 mmol) of magnesium, and a catalytic amount of iodine were suspended in 100 mL of tetrahydrofuran and refluxed for 3 hours to generate a Grignard reagent. 8.29 g (38 mmol) of Intermediate [14-A] was further added to the mixture and refluxed for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 10.6 g (28.5 mmol) of Intermediate [14-B].

(266) 3) Synthesis of Intermediate [14-C]

(267) 10.6 g (28.5 mmol) of Intermediate [14-B] was added to an excess of hydrochloric acid/acetic acid solution and refluxed for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.59 g (24.2 mmol) of Intermediate [14-C].

(268) 4) Synthesis of Intermediate [14-D]

(269) 8.59 g (24.2 mmol) of Intermediate [14-C] and 1.45 g (121 mmol) of imidazole were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 7.61 g (9.68 mmol) of Intermediate [14-D].

(270) 5) Synthesis of Intermediate [14-E]

(271) 7.61 g (9.68 mmol) of Intermediate [14-D] and 4.72 g (29 mmol) of ammonium hexafluorophosphate were added to a reaction container and suspended in 20 mL of methyl alcohol and 20 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 6.37 g (7.74 mmol) of Intermediate [14-E].

(272) 6) Synthesis of Compound 14

(273) 6.37 g (7.74 mmol) of Intermediate [14-E], 3.2 g (8.51 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 1.92 g (23.2 mmol) of sodium acetate were suspended in 90 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110° C. for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 1.34 g (1.86 mmol) of Compound 14.

Synthesis Example 4

Synthesis of Compound 19

(274) ##STR00091## ##STR00092##

(275) Compound 19 was obtained in the same manner as in Synthesis Example 2, except that dichloro(2,4,6-trimethylphenyl)borane was used instead of dimethyldichlorosilane.

Synthesis Example 5

Synthesis of Compound 24

(276) ##STR00093##

(277) Compound 24 was obtained in the same manner as in Synthesis Example 1, except that 4,5-dimethyl-1H-imidazole was used instead of imidazole.

(278) ¹H NMR and MS/FAB of the compounds synthesized in Synthesis Examples 1 to 5 are shown in Table 1.

(279) Synthesis methods of compounds other than Compounds shown in Table 1 may also be easily recognized by those of skill in the art by referring to the synthesis mechanisms and methods described above.

(280) TABLE-US-00001 TABLE 1 MS/FAB Compound ¹H NMR (CDCl₃, 400 MHz) found calc. 4 0.82-1.81 (8H, m), 2.62 (2H, m), 6.45-6.55 (6H, m), 575.1290 575.1285 7.10 (2H, m), 8.55 (2H, m) 9 0.25 (6H, s), 1.11-1.89 (8H, m), 2.52 (2H, m), 5.83 617.1584 617.1574 (1H, m), 6.31 (2H, m), 6.52 (2H, m), 6.81 (2H, m), 7.22 (1H, m), 8.74 (2H, m) 14 1.1-1.98 (8H, m), 2.65 (2H, m), 6.57 (2H, m), 6.71- 723.1953 723.1962 6.82 (4H, m), 7.09-7.48 (8H, m), 7.91 (2H, m), 8.75 (2H, m) 19 0.81-1.91 (8H, m), 2.18 (3H, s), 2.41 (6H, s), 2.42 689.2285 689.2290 (2H, m), 6.46 (2H, m), 6.81 (2H, m), 7.01 (2H, m), 7.24-7.65 (4H, m), 8.51 (2H, m) 24 0.84-1.85 (8H, m), 2.31 (12H, s) 2.63 (2H, m), 6.41- 631.1910 631.1911 6.45 (4H, m), 7.12 (2H, m)

EXAMPLES

Example 1

(281) As a substrate and an ITO anode, a Corning 15 Ω/cm² (1,200 Å) ITO glass substrate was cut to a size of 50 mm×50 mm×0.7 mm, sonicated with isopropyl alcohol and pure water each for 5 minutes, and then cleaned by exposure to ultraviolet rays and ozone for 30 minutes. Then, the ITO glass substrate was provided to a vacuum deposition apparatus.

(282) With reference to the compounds illustrated below, 2-TNATA was vacuum-deposited on the ITO anode formed on the ITO glass substrate to form a hole injection layer having a thickness of 600 Å, and NPB was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 300 Å.

(283) BCPDS and POPCPA (BCPDS to POPCPA weight ratio of 1:1) as a co-host and Compound 4 as a dopant were co-deposited on the hole transport layer as a co-host to dopant weight ratio of 90:10 to form an emission layer having a thickness of 300 Å.

(284) TSPO1 was deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å, Alga was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, LiF was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited on the electron injection layer to form a cathode having a thickness of 3,000 Å, thereby completing the manufacture of an organic light-emitting device.

(285) ##STR00094## ##STR00095##

Examples 2 to 5 and Comparative Examples 1 to 4

(286) Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds shown in Table 2 were each used instead of Compound 1 as a dopant in forming an emission layer.

(287) ##STR00096##

Evaluation Example 1

(288) The driving voltage, current density, brightness, luminance efficiency, emission color, and wavelength of maximum emission of the organic light-emitting devices manufactured according to Examples 1 to 5 and Comparative Examples 1 to 4 were measured by using Keithley SMU 236 and a luminance meter PR650, and results thereof are shown in Table 2.

(289) TABLE-US-00002 TABLE 2 wavelength of Driving Current Luminance maximum Dopant voltage density Brightness efficiency Emission emission compound (V) (mA/cm²) (cd/m²) (cd/A) color (nm) Example 1 Compound 4 5.30 50 4125 8.25 Blue 490 Example 2

Compound 9 5.45 50 4025 8.05 Blue 470 Example 3 Compound 14 5.55 50 4033 8.07 Blue 474
 Example 4 Compound 19 5.43 50 4024 8.05 Blue 468 Example 5 Compound 24 5.58 50 4055 8.11
 Blue 495 Comparative Example 1 6.56 50 3870 7.74 Blue 478 Example 1 Comparative Compound A
 7.10 50 2957 6.89 Blue 460 Example 2 Comparative Compound B 6.83 50 3800 7.34 Blue 500
 Example 3 Comparative Compound C 6.23 50 3759 7.12 Blue 480 Example 4
 (290) Referring to Table 2, it can be seen that the organic light-emitting devices of Examples 1 to 5
 had a low driving voltage and high brightness and luminance efficiency, as compared with those of
 the organic light-emitting devices of Comparative Examples 1 to 4.
 (291) As described above, an organic light-emitting device including the organometallic compound
 may have a low driving voltage, high brightness, and high luminance efficiency.
 (292) Example embodiments have been disclosed herein, and although specific terms are
 employed, they are used and are to be interpreted in a generic and descriptive sense only and not
 for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art
 as of the filing of the present application, features, characteristics, and/or elements described in
 connection with a particular embodiment may be used singly or in combination with features,
 characteristics, and/or elements described in connection with other embodiments unless otherwise
 specifically indicated. Accordingly, it will be understood by those of skill in the art that various
 changes in form and details may be made without departing from the spirit and scope of the present
 invention as set forth in the claims.

Claims

1. An organic light-emitting device comprising: a first electrode; a second electrode; and an organic layer between the first electrode and the second electrode and comprising an emission layer, wherein the organic layer comprises an organometallic compound represented by Formula 1: ##STR00097## wherein, in Formula 1, M.sub.11 is selected from platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm), Y.sub.10, Y.sub.20, Y.sub.30, and Y.sub.40 are each independently N or C, Y.sub.11, Y.sub.12, Y.sub.21, Y.sub.22, Y.sub.31, Y.sub.32, Y.sub.41, and Y.sub.42 are each independently N or C, A.sub.30, and A.sub.40 are each independently selected from a C.sub.5-C.sub.60 carbocyclic group and a C.sub.1-C.sub.60 heterocyclic group, T.sub.11 to T.sub.14 are each independently selected from a single bond, *—O—*, and *—S—*, L.sub.11 is selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, a₁₁ is selected from 1, 2, 3, 4, and 5, at least one selected from L.sub.11 (s) in the number of a₁₁ is a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, L.sub.12 and L.sub.14 are each independently selected from a single bond, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, L.sub.13 is selected from a single bond, *—O—*, *—S—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, wherein, if L.sub.11 is a cyclohexylene group, L.sub.13 is *—B(R.sub.1)—*, a₁₂, and a₁₄ are each independently selected from 0, 1, 2, 3, 4, and 5, a₁₃ is selected from 1, 2, 3, 4, and 5, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted

C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted C.sub.1-C.sub.60 hetero aryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 hetero arylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —N(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), and —P(=S)(Q.sub.1)(Q.sub.2), R.sub.1 and R.sub.2 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, R.sub.1 and R.sub.10; R.sub.1 and R.sub.20; R.sub.1 and R.sub.30; or R.sub.1 and R.sub.40 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b10, b20, b30, and b40 are each independently selected from 1, 2, 3, 4, 5, 6, 7, and 8, * and *' each indicate a binding site to a neighboring atom, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is selected from: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic

group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 hetero aryloxy group, a C.sub.1-C.sub.60 hetero arylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C.sub.1-C.sub.60 alkyl group substituted with at least one selected from deuterium, —F, and a cyano group, a C.sub.6-C.sub.60 aryl group substituted with at least one selected from deuterium, —F, and a cyano group, a biphenyl group, and a terphenyl group, and A.sub.10 and A.sub.20 are each independently selected from groups represented by Formulae 10-1 to 10-3: ##STR00098## wherein, in Formulae 10-1 to 10-3, Y.sub.13 to Y.sub.18 and X.sub.11 to X.sub.14 are each independently N or C(R.sub.10), and *, *, and *' each indicate a binding site to a neighboring atom.

2. The organic light-emitting device of claim 1, wherein the first electrode is an anode, the second electrode is a cathode, the organic layer further comprises a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, the hole transport region comprises at least one layer selected from a hole injection layer, a hole transport layer, a buffer layer, an emission auxiliary layer, and an electron blocking layer, and the electron transport region comprises at least one layer selected from a hole blocking layer, an electron transport layer, and an electron injection layer.

3. The organic light-emitting device of claim 2, wherein the emission layer comprises the organometallic compound.

4. The organic light-emitting device of claim 1, wherein the emission layer emits blue light having a maximum emission wavelength of about 440 nm to about 500 nm.

5. The organic light-emitting device of claim 3, wherein the emission layer comprises a host and a dopant, and the dopant comprises the organometallic compound.

6. The organic light-emitting device of claim 3, wherein the emission layer further comprises a host, and the host comprises a silyl-containing compound, a phosphine oxide-containing compound, or a combination thereof.

7. The organic light-emitting device of claim 2, wherein the electron transport region comprises a hole blocking layer, the hole blocking layer directly contacts the emission layer, and the hole blocking layer comprises a phosphine oxide-containing compound or a silyl-containing compound.

8. An organometallic compound represented by Formula 1: ##STR00099## wherein M.sub.11 is selected from platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium

(Eu), terbium (Tb), and thulium (Tm), Y.sub.10, Y.sub.20, Y.sub.30, and Y.sub.40 are each independently N or C, Y.sub.11, Y.sub.12, Y.sub.21, Y.sub.22, Y.sub.31, Y.sub.32, Y.sub.41, and Y.sub.42 are each independently N or C, A.sub.30, and A.sub.40 are each independently selected from a C.sub.5-C.sub.60 carbocyclic group and a C.sub.1-C.sub.60 heterocyclic group, T.sub.11 to T.sub.14 are each independently selected from a single bond, *—O—*, and *—S—*, L.sub.11 is selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, a₁₁ is selected from 1, 2, 3, 4, and 5, at least one selected from L.sub.11 (s) in the number of a₁₁ is a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, L.sub.12 and L.sub.14 are each independently selected from a single bond, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, L.sub.13 is selected from a single bond, *—O—*, *—S—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—P(R.sub.1)(R.sub.2)—*, and *—Ge(R.sub.1)(R.sub.2)—*, wherein, if L.sub.11 is a cyclohexylene group, L.sub.13 is *—B(R.sub.1)—*, a₁₂, and a₁₄ are each independently selected from 0, 1, 2, 3, 4, and 5, a₁₃ is selected from 1, 2, 3, 4, and 5, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted C.sub.1-C.sub.60 hetero aryloxy group, a substituted or unsubstituted C.sub.1-C.sub.60 hetero arylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —N(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), and —P(=S)(Q.sub.1)(Q.sub.2), R.sub.1 and R.sub.2 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, R.sub.1 and R.sub.10; R.sub.1 and R.sub.20; R.sub.1 and R.sub.30; or R.sub.1 and R.sub.40 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b₁₀, b₂₀, b₃₀, and b₄₀ are each independently selected from 1, 2, 3, 4, 5, 6, 7, and 8, * and *' each indicate a binding site to a neighboring atom, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is selected from: deuterium, —F, —Cl, —Br, —I,

a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a C.sub.1-C.sub.60 hetero aryloxy group, a C.sub.1-C.sub.60 hetero arylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C.sub.1-C.sub.60 alkyl group substituted with at least one selected from deuterium, —F, and a cyano group, a C.sub.6—Coo aryl group substituted with at least one selected from deuterium, —F, and a cyano group, a biphenyl group, and a terphenyl group, and A.sub.10 and A.sub.20 are each independently selected from groups represented by Formulae 10-1 to 10-3: ##STR00100## wherein, in Formulae 10-1 to 10-3, Y.sub.13 to Y.sub.18 and X.sub.11 to X.sub.14 are each independently N or C(R.sub.10), and *, *' and *'' each indicate a binding site to a neighboring atom.

9. The organometallic compound of claim 8, wherein M.sub.11 is selected from Pt, Pd, Cu, Ag, and

Au.

10. The organometallic compound of claim 8, wherein at least one selected from Y.sub.10 and Y.sub.20 is C, (i) when Y.sub.10 is C, a bond between Y.sub.10 and T.sub.11 or a bond between Y.sub.10 and M.sub.11 is a coordinate bond, and (ii) when Y.sub.20 is C, a bond between Y.sub.20 and T.sub.12 or a bond between Y.sub.20 and M.sub.11 is a coordinate bond.

11. The organometallic compound of claim 8, wherein A.sub.30 and A.sub.40 are each independently selected from groups represented by Formulae 2-1 to 2-43: ##STR00101##
##STR00102## ##STR00103## ##STR00104## wherein, in Formulae 2-1 to 2-43, X.sub.21 to X.sub.23 are each independently selected from C(R.sub.24) and C—*, and at least two selected from X.sub.21 to X.sub.23 are each independently C—*, X.sub.24 is N—*, and X.sub.25 and X.sub.26 are each independently selected from C(R.sub.24) and C—*, wherein at least one selected from X.sub.25 and X.sub.26 is C—*, X.sub.27 and X.sub.28 are each independently selected from N, N(R.sub.25), and N—*, and X.sub.29 is selected from C(R.sub.24) and C—*, wherein i) at least one selected from X.sub.27 and X.sub.28 is N—*, and X.sub.29 is C—*, or ii) X.sub.27 and X.sub.28 are each independently N—*, and X.sub.29 is C(R.sub.24), R.sub.21 to R.sub.25 are each independently the same as described in connection with R.sub.10 in claim 8, b21 is selected from 1, 2, and 3, b22 is selected from 1, 2, 3, 4, and 5, b23 is selected from 1, 2, 3, and 4, b24 is selected from 1 and 2, and * indicates a binding site to a neighboring atom.

12. The organometallic compound of claim 8, wherein each of T.sub.11 to T.sub.14 is a single bond; T.sub.11 is selected from O and S, and T.sub.12 to T.sub.14 are each independently a single bond; T.sub.12 is selected from O and S, and T.sub.11, T.sub.13, and T.sub.14 are each independently a single bond; T.sub.13 is selected from O and S, and T.sub.11, T.sub.12, and T.sub.14 are each independently a single bond; or T.sub.14 is selected from O and S, and T.sub.11, T.sub.12, and T.sub.13 are each independently a single bond.

13. The organometallic compound of claim 8, wherein at least one selected from L.sub.11 (s) in the number of a11 is selected from: a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, and a cyclodecanylene group; and a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, and a cyclodecanylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkenyl group, a C.sub.1-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

14. The organometallic compound of claim 8, wherein at least one selected from L.sub.11 (s) in the number of a11 is selected from groups represented by Formulae 3-1 to 3-5: ##STR00105## wherein, in Formulae 3-1 to 3-5, R.sub.3 is selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkenyl group, a C.sub.1-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, b4 is an integer from 1 to 4, b6 is an integer from 1 to 6, b8 is an integer from 1 to 8, b10 is an integer from 1 to 10, b12 is an integer from 1 to 12, and * and *' each indicate a binding site to a neighboring atom.

15. The organometallic compound of claim 8, wherein L.sub.12 and L.sub.14 are each

independently a single bond, and L.sub.13 is selected from *—O—*, *—S—*, *—N(R.sub.1)—*, and *—B(R.sub.1)—*.

16. The organometallic compound of claim 8, wherein R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 are each independently selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group; a group represented by one selected from Formulae 5-1 to 5-25; and a group represented by one selected from Formulae 6-1 to 6-55, and R.sub.1 and R.sub.2 are optionally linked to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, or a fluorene group: ##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112## ##STR00113## wherein, in Formulae 5-1 to 5-25 and 6-1 to 6-55, Y.sub.31 is O, S, C(Z.sub.34)(Z.sub.35), N(Z.sub.34), or Si(Z.sub.34)(Z.sub.35), Z.sub.31 to Z.sub.35 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkenyl group, a C.sub.1-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group, e2 is 1 or 2, e3 is an integer from 1 to 3, e4 is an integer from 1 to 4, e5 is an integer from 1 to 5, e6 is an integer from 1 to 6, e7 is an integer from 1 to 7, e9 is an integer from 1 to 9, and * indicates a binding site to a neighboring atom.

17. The organometallic compound of claim 8, wherein R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, and R.sub.40 are each independently selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, and a dibenzocarbazolyl group; and a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, and a dibenzocarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, and a biphenyl group, and R.sub.1 and R.sub.2 are optionally linked to form a cyclopentane group, a cyclohexane group, or a fluorene group.

18. An organometallic compound, wherein the organometallic compound is selected from Compounds 1 to 3 and 5 to 20: ##STR00114##
